

Q1. Student Grade Book Application

CODE

```
students = {}

for i in range(5):
    name = input()
    marks = list(map(int, input().split()))
    students[name] = marks

def get_grade(avg):
    if avg ≥ 90:
        return "O"
    elif avg ≥ 80:
        return "A+"
    elif avg ≥ 70:
        return "A"
    elif avg ≥ 60:
        return "B+"
    elif avg ≥ 50:
        return "B"
    else:
        return "F"

result = []

# Calculate total, average, and grade
for name, marks in students.items():
    total = sum(marks)
    avg = total / 5
    result.append((name, total, avg, get_grade(avg)))

# Sort by average in descending order
result.sort(key=lambda x: x[2], reverse=True)

# Find topper
topper = result[0][0]

# Find subject with highest average
subject_avg = []
for i in range(5):
    s = 0
    for marks in students.values():
        s += marks[i]
    subject_avg.append(s / len(students))

best_subject = subject_avg.index(max(subject_avg)) + 1

# Display result
print("Name\tTotal\tAverage\tGrade")
for r in result:
    print(f"{r[0]}\t{r[1]}\t{r[2]:.2f}\t{r[3]}")

print("Topper:", topper)
print("Best Subject: Subject", best_subject)
```

INPUT

```
Ravi
90 85 80 95 88
Rahul
70 75 80 65 60
Anu
95 92 90 94 96
Priya
60 65 70 75 80
Kiran
50 55 60 58 62
```

OUTPUT

```
(no output)
```

Q2. Debugging and Rewriting

CODE

```
accounts = {}

def create_account(acc_no, name, balance):
    accounts[acc_no] = {'name': name, 'balance': balance, 'txns': []}

def deposit(acc_no, amount):
    if acc_no in accounts:
        accounts[acc_no]['balance'] += amount
        accounts[acc_no]['txns'].append(("Deposit", amount))

def withdraw(acc_no, amount):
    if accounts[acc_no]['balance'] >= amount:
        accounts[acc_no]['balance'] -= amount
        accounts[acc_no]['txns'].append(("Withdraw", amount))
    else:
        print("Insufficient balance")

def mini_statement(acc_no):
    for t in accounts[acc_no]['txns'][-5:]:
        print(t)

create_account(1001, "Ravi", 10000)
deposit(1001, 5000)
withdraw(1001, 20000)
withdraw(1001, 3000)

print("Balance:", accounts[1001]['balance'])
print("Mini Statement:")
mini_statement(1001)
```

OUTPUT

(no output)